

The FLUX Pipeline

FLUX: a pipeline for MEG and OPM analysis



Aim

Magnetoencephalography (MEG), based on SQUID (superconducting quantum interference devices) and OPM (optically pumped magnetometer) sensors, enables the non-invasive measurement of human brain activity with millisecond temporal resolution. In addition to capturing rapid neural dynamics, MEG allows for the estimation of the spatial origin of neural signals. These capabilities make MEG a powerful tool in cognitive neuroscience, though its full potential depends on advanced signal processing and source modelling techniques.

To support such analyses, the research community has developed several open-source toolboxes. While these toolboxes offer extensive analytical flexibility, this very flexibility can pose challenges—particularly in ensuring reproducible research and in supporting researchers new to the field. The FLUX pipeline addresses these challenges by making analysis steps and parameter settings explicit for standard procedures in cognitive neuroscience.

The FLUX pipeline originates from the Cogitate Consortium, which investigates specific questions in cognitive neuroscience. It is implemented in MNE-Python (a Python-based toolbox), using a dataset on visuospatial attention to demonstrate the full analysis workflow. An earlier version was also implemented in FieldTrip (a MATLAB-based toolbox) to ensure consistency across platforms; however, the FieldTrip version is no longer maintained.

The pipeline is delivered through Jupyter Notebooks and MATLAB's Live Editor, combining code, graphical outputs, and detailed explanations and justifications for each analysis step. In addition, it provides suggested text and parameter settings for use in registrations and publications—enhancing reproducibility and supporting pre-registration practices.

Designed to support both self-guided learning and instructor-led workshops, the FLUX pipeline also serves as an educational resource. Our goal is to strengthen the MEG community by introducing a degree of standardisation to foundational analysis steps and harmonising approaches across different toolboxes. We also aim to support newcomers to the field through accessible training materials.

Importantly, the FLUX pipeline is a living resource—it will continue to evolve in step with toolbox developments and emerging

insights.

This first version of the FLUX Pipeline is published in Ferrante et al. (2022) Neuroimage



Neuronal Oscillations



Watch on

Target users

The target audience includes researchers who are new to MEG and OPM research, as well as more experienced users looking

The Neuronal Oscillations Group



The data sets

The dataset for the FLUX pipeline based on MNE-Python is available on OpenNeuro and Zoneto and is used throughout the step-by-step tutorials. We provide datasets from a MEGIN MEG system, a Cereba/QuSpin OPM system and well as a FieldLine OPM system. The data originates from a simple spatial attention paradigm in which participants are instructed to attend either left or right to a moving grating. This task and stimulation reliably elicit modulations in the alpha and gamma frequency bands.

Standard Operation Procedure (SOP) for MEGIN MEG data acquisition

We here share an example of a SOP for MEG data acquisition. It is derived from the Cogitate project and relevant parts can be adapted to specific studies and used for preregistrations.

Standard Operation Procedure (SOP) for Cereba OPM data acquisition

Under developments

The FLUX pipeline for MEGIN MEG data

The pipeline is implemented as Jupyter scripts maintained on a [GitHub server](#)

1. Introduction and installation
2. The dataset
3. A first look at the data
4. Artefact attenuation by MaxFilter
5. Artefacts annotations
6. ICA for attenuating artefacts
7. Extracting condition-specific trials
8. Event-related fields
9. Time-frequency representations of power
10. Classification using multivariate patterns analysis
11. Constructing the forward model
12. Source modelling using DICS beamforming

The FLUX pipeline for Cerca/QuSpin OPM data

New paradigm to be used for July 2026 toolkit: download from Zonedo.
 BACKUP (old paradigm) TO BE USED IN OPM FLUX TOOLKIT JULY 2026 AT OXFORD

The spatial attention data used on website to download from Zonedo.

The pipeline is implemented as Jupyter scripts maintained on a [GitHub server](#). It has been test for MNE Python v. 1.12.1

1. Introduction and installation
2. The dataset
3. Down-sampling and BIDS conversion
4. A first look at the data
5. Artefact attenuation by HFC
6. Annotating muscle and ocular artefacts
7. ICA for attenuating artefacts
8. Extracting trials specific conditions
9. Event Related Fields
10. Time-frequency representations of power
11. Classification based on multi-variate pattern analysis
12. Constructing the forward model
13. Source modelling using using DICS beamforming

The FLUX pipeline for FieldLine OPM data

TO BE USED IN OPM FLUX TOOLKIT OCTOBER 2025 IN BERLIN
 It has been test for MNE Python v. 1.9.0

The spatial attention data to download from Zonedo

The pipeline is implemented as Jupyter scripts maintained on a GitHub server

The pipeline is implemented as Jupyter scripts maintained on a GitHub server

1. Introduction and installation
2. The dataset
3. Down-sampling and BIDS conversion
4. A first look at the data
5. Artefact attenuation by HFC
6. Annotating muscle and ocular artefacts
7. ICA for attenuating artefacts
8. Extracting trials specific conditions
9. Event Related Fields
10. Time-frequency representations of power
11. Classification based on multi-variate pattern analysis
12. Head digitisation transformations (FieldLine specific)
13. Constructing the forward model
14. Source modelling using using DICS beamforming

The FLUX pipeline – FieldTrip

Note, that we are no longer maintaining the FLUX pipeline for Matlab/FieldTrip. The scripts here are meant to support users that are in the process of transitioning from FieldTrip to MNE Python.

The pipeline is implemented as Matlab Livescripts scripts on a GitHub server.

1. Introduction and installation
2. The dataset
3. A first look at the data
4. Artefact attenuation by MaxFilter
5. Extracting condition-specific trials
6. Semi-automatic artefact rejection
7. ICA for attenuating artefacts
8. Event-related fields
9. Time-frequency representations of power
10. Constructing the forward model
11. Source modelling using DICS beamforming